Machine Learning Algorithms for Credit Card Fraud

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Dataset: https://www.kaggle.com/datasets/kaushalnandania/credit-card-fraud-detection

Code: https://github.com/kjelstromjr/Credit-Fraud

Project Description and Hypothesis

Credit card fraud has been an issue as long as the cards have existed. After the transition to handling credit card transitions electronically, it has become easier for criminals to obtain card information and use it to steal money. The transition to digital also means that companies have access to large amounts of data that can be used to prevent fraud. My goal for this project is to find a model that accurately predicts whether or not a credit card transaction is fraud or not. I hypothesize that the location of the transaction and the charge amount will have the most impact on the decision of the models.

Dataset

The dataset I chose to use for this project can be found on Kaggle. It's a synthetic dataset posted by Kaushal Nandaniya created using the Sparkov Data Generation Tool by Brandon Harris. The dataset contains 1.85 million simulated transactions with 23 features. The columns for the data include information like time/date, amount, location, merchant, category customer info and card number. Below is some example data from the testing file. I've put the row into columns for visual representation on a page:

Feature Name	1	2	3
trans_date_trans_time	6/21/2020 12:14:25 PM	6/21/2020 12:14:33 PM	6/21/2020 12:14:53 PM
cc_num	2291163933867244	3573030041201292	3598215285024754
merchant	fraud_Kirlin and Sons	fraud_Sporer-Keebler	fraud_Swaniawski, Nitzsche and Welch
category	personal_care	personal_care	health_fitness
amt	2.86	29.84	41.28
first	Jeff	Joanne	Ashley
last	Elliott	Williams	Lopez
gender	M	F	F
street	351 Darlene Green	3638 Marsh Union	9333 Valentine Point
city	Columbia	Altonah	Bellmore
state	SC	UT	NY
zip	29209	84002	11710
lat	33.9659	40.3207	40.6729
long	-80.9355	-110.436	-73.5365
job	Mechanical engineer	Sales professional, IT	Librarian, public
dob	3/19/1968	1/17/1990	10/21/1970
merchant_lat	33.986391	39.450497999999996	40.49581
merchant_long	81.200714	-109.960431	-74.196111
is_fraud	0	0	0

The data in this set is extremely imbalanced, which accurately represents the data that would be collected in the real world. This introduces an interesting challenge since the models should be able to identify the majority of test cases. This however is not what I am looking for since the models need to be able to identify the small number of cases of fraud.

Data Preprocessing

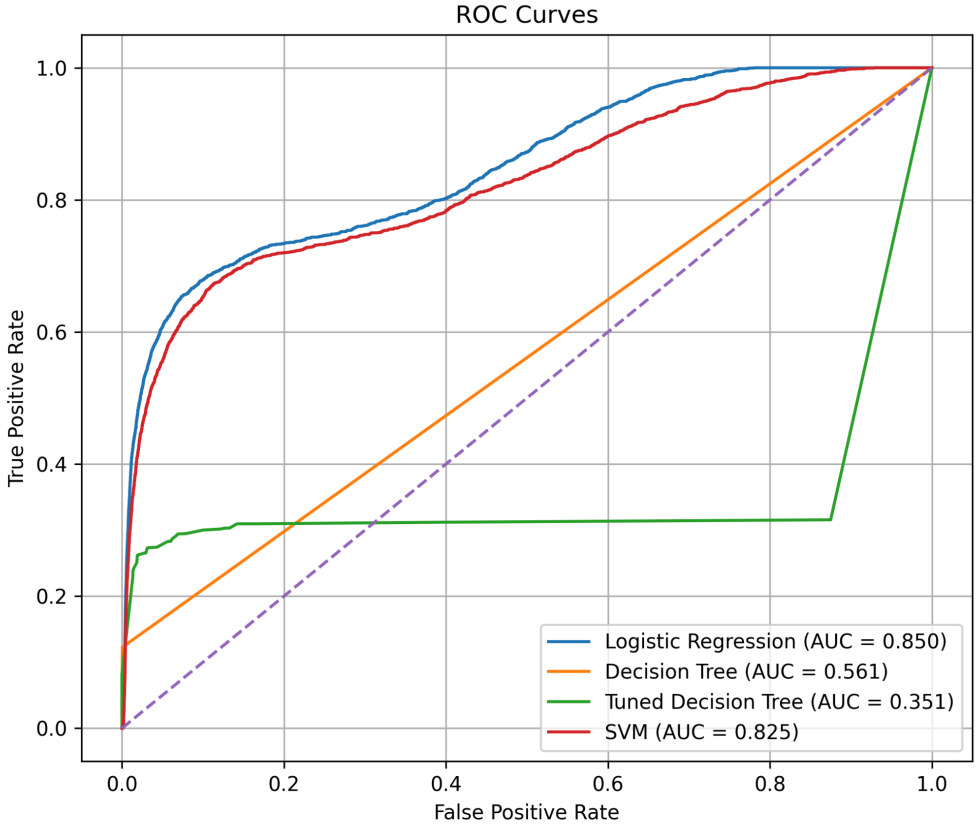
This dataset contains a lot of strings. So there needs to be a good amount of data preprocessing. There are three main things that I did to the data to prepare it for training: one-hot encoding, target encoding, and some feature engineering. My initial thought for this project was to use one-hot encoding for all of the string features, but after testing, I decided not to due to the immense amount of memory and time required for training. However, for some of the features with less variance like category, merchant, and gender, I decided to use one-hot encoding. Other features with more variance like city, merchant, and job were converted using target encoding. Some of the number features also needed some engineering. For any time variable, I decided to split it up into the hour, day, month, and day of week. This made it so that each of the features were just one number. I also decided to change the date of birth into a single number named age. The rest of the features: CC number, name, transaction number, and street were removed since they don’t add any additional value to the purpose of this project. After the cleaning and engineering, all of the kept data features were standardized since there is a lot of variance in the size of numbers used.

Model Implementations

All of the models for this project were implemented using the sklearn python library. I chose to test and compare four different models with three different algorithms for this project: logistic regress, decision tree, tuned decision tree, and SVM. Each of these have their own strengths and weaknesses, but I believe that they all have the potential to perform great in this classification problem. Each of the models were trained using the same training data and are tested with the same testing data. The tuned decision tree differs from the regular decision tree in that the max depth is set to 8 and the minimum number of samples per leaf is set to 20. I will also explain a little bit more into why I decided to keep in the tuned decision tree in the results.

Evaluation and Results

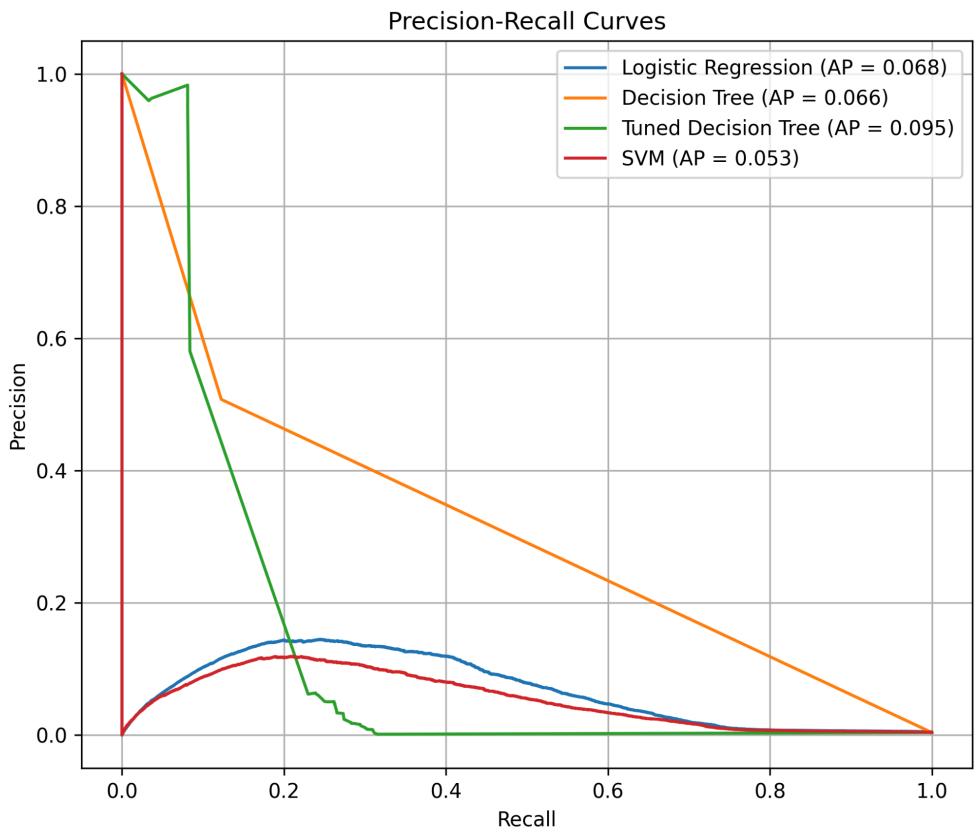
To test and compare each of the models, I decided to use a couple different metrics. I first wanted to view the performance of each of the models across the entire dataset via ROC curves. This gives an idea about which models perform best when it comes to simply recognizing that a transaction is not fraudulent. Below is a graph of each of the model's ROC curves:



In the graph, it's clear to see that the logistic regression and SVM models performed much better for the majority of the data. Both models show normal ROC curves which indicates that models learned well. The decision tree lines show a different story. Both show little to no curvature. This might suggest that the models were overfitting to the data. This is especially true for the untuned decision tree which is close to the dotted line indicating where a random state would be. Another thing to point out here is how the tuned decision tree is below the random line meaning it significantly performed poorly.

While the data presented by the ROC curves is helpful, it's only a fraction of the whole picture. Like mentioned previously, the information in this dataset is extremely imbalanced. Thus, the ROC curve is not the best indicator for whether or not a model is

able to identify fraudulent activity. This leads into the second set of measurements I chose to compare. Precision and recall are both great metrics for this test case, especially since they have to do with the accuracy of the models. This means that they can show how the models are performing for the small number of transactions that are fraud. Below is a graph showing the precision-recall curves of each of the trained models:



The results change a bit when looking at the precision-recall curves. At first glance, the decision trees appear to perform better with the majority of their lines being higher on the graph when compared to logistic regression and SVM. However, the same issues as shown in the ROC curves appear to transfer to this graph as well. Both the regular and tuned decision trees have very jagged edges and have dramatic drops. In summary, the

decision trees performed more like linear functions in this case, which is not ideal. On the other side, the logistic regression and SVM models maintained their curve structure in the graph. However, both models were on the lower end of performance when it came to actually finding the fraud cases.

Along with measuring and plotting a few metrics, I also collected data on which features had the most impact for each of the models. Below is a table showcasing the top 3 most impactful features for each model along with their coefficient (logistic regression and SVM) and importance (decision trees) scores:

Logistic Regression	**Decision Tree**	**Tuned Decision Tree**	**SVM**
Amount (1.67)	Amount (0.68)	Amount (0.73)	Amount (0.34)
Category (shopping_net) (1.28)	Category (gas_transport) (0.08)	Category (gas_transport) (0.08)	Zip code (0.24)
Category (shopping_pos) (0.88)	Hour (0.08)	Hour (0.08)	Category (shopping_net) (0.24)

Analysis and Discussion

Overall, I thought the results of these models were both surprising and intriguing. It was interesting to see how big of an impact the imbalanced data had on the training. Specifically, the decision trees had a hard time actually learning the overall patterns. I think these results are a great example showing that decision trees are not the best when it comes to large imbalances. Using the data from the important features, I think that we actually had another thing that had a huge impact on the decision trees not performing well. In both instances, the trees largely favored the amount of the

transaction over all other features. I think this became an issue simply due to the way the algorithms work. The algorithm almost behaves like a shortest path algorithm in that it finds the quickest path to an answer. So I think the algorithm got caught in the instances with large transaction amounts, which would have a larger chance of being fraud. This would also mean that the decision trees would have missed any other patterns.

I believe that the logistic regression and SVM were more consistently able to identify these less noticeable patterns. This makes sense too since these algorithms are more focused on finding a boundary rather than a shortest path. The fact that these algorithms performed better overall also might suggest that the data is already mostly linear. This would make sense with the decision tree too since that algorithm would have a hard time finding linear relationships. Another reason why the logistic regression and SVM algorithm might have performed better was simply because of how imbalanced the data is. Both of these algorithms have some sort of regularization that can help in cases like this to prevent overfitting. We can actually see this pretty well in both graphs. The decision trees are suggesting overfitting with their jagged edges and inconsistency, whereas the logistic regression and SVM models show curvature and consistent results.

I also found it interesting how the translation amount feature was the most important across all models. It’s also interesting that, on average, category was the second most important feature. At least for the amount, I find this to be very realistic. Large transactions for the average person would be out of the ordinary and I wouldn’t consider an algorithm wrong for flagging a transaction like that as fraud.

Conclusion

This project has revealed some interesting results. To summarize what I have learned: decision trees are not ideal for imbalanced datasets and that the amount of a transaction has the most impact on whether or not a transaction is fraud. If I had to choose a winning model, I would choose logistic regression. Despite its lower performance when it comes to precision and recall, I think it has a lot of potential to be a near perfect model for this dataset. Since the model performs very well with the majority of the dataset (ROC), I think that the model can be improved with some tuning and maybe some more feature engineering. Ideally, I would want to get the model to have a precision-recall curve that is much higher on the graph. It might also be interesting to look into algorithms like XGBoost which has the possibility of performing better in an imbalanced scenario. Credit card fraud is still a very real and very dangerous threat in our world that needs to be handled properly and effectively. I believe that my data collected shows that machine learning algorithms are key in catching these types of threats. I don’t have access to the data that large organizations like banks and credit card companies have and my models were able to identify realistic patterns in everyday transactions. Imagine the power and accuracy a model would have with even more features and data points.